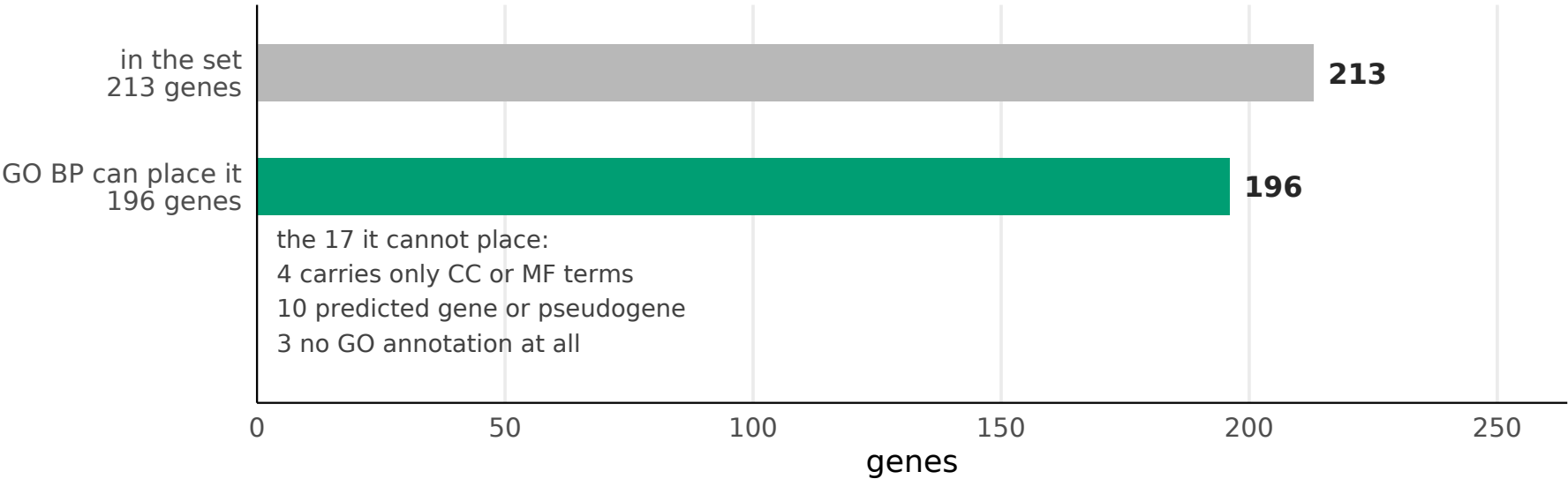


What the mouse 39 °C-derived up arm is made of, asked of the ontology

GO BP semantic similarity over the 213 gene set, read instead of the expression (Wang measure, BMA combiner, GOSemSim 2.39.2, GO.db 3.22.0). Nothing here reads expression, so no grouping shown can have been derived from an enrichment result. Semantic proximity means two genes carry similar annotation. It is not evidence of co-regulation, and an enriching set is not evidence that the process it is named for is present.

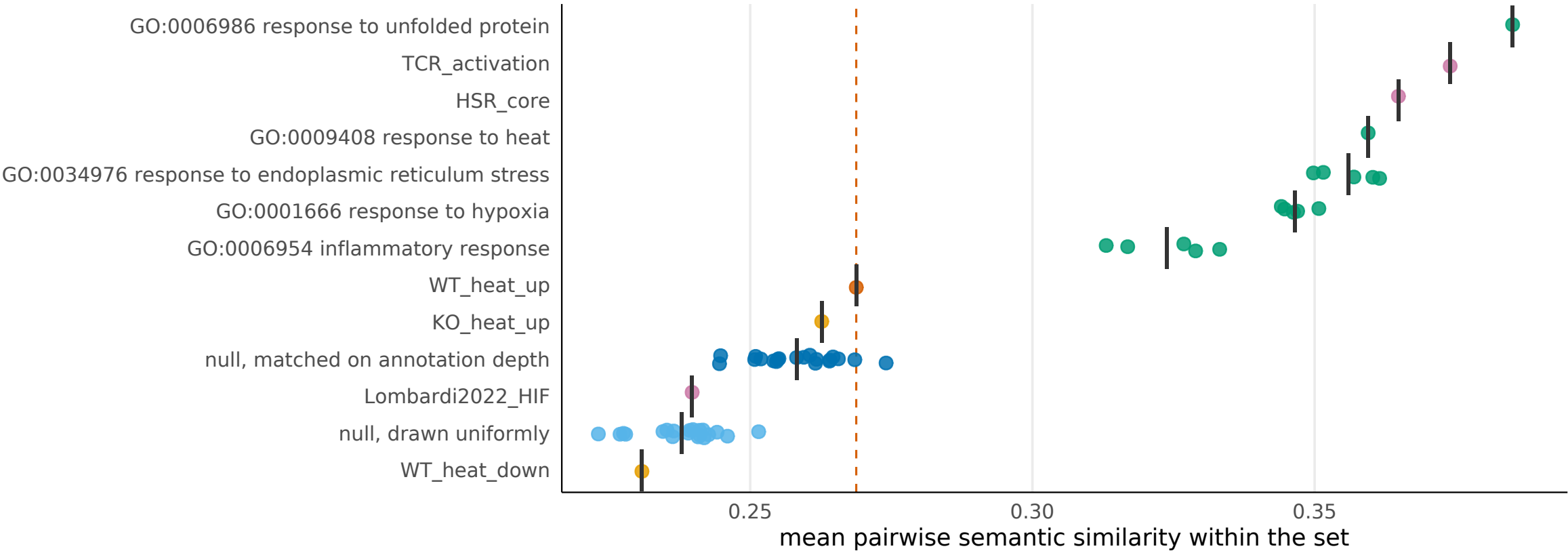
Can GO BP see the set?

Curated evidence only, with IEA annotation set aside (keeping it would recover 1). The gap is not neutral: P4ha2 carries no BP term in mouse or in human, so a lens overlap drawn elsewhere contains a gene this panel has to drop.



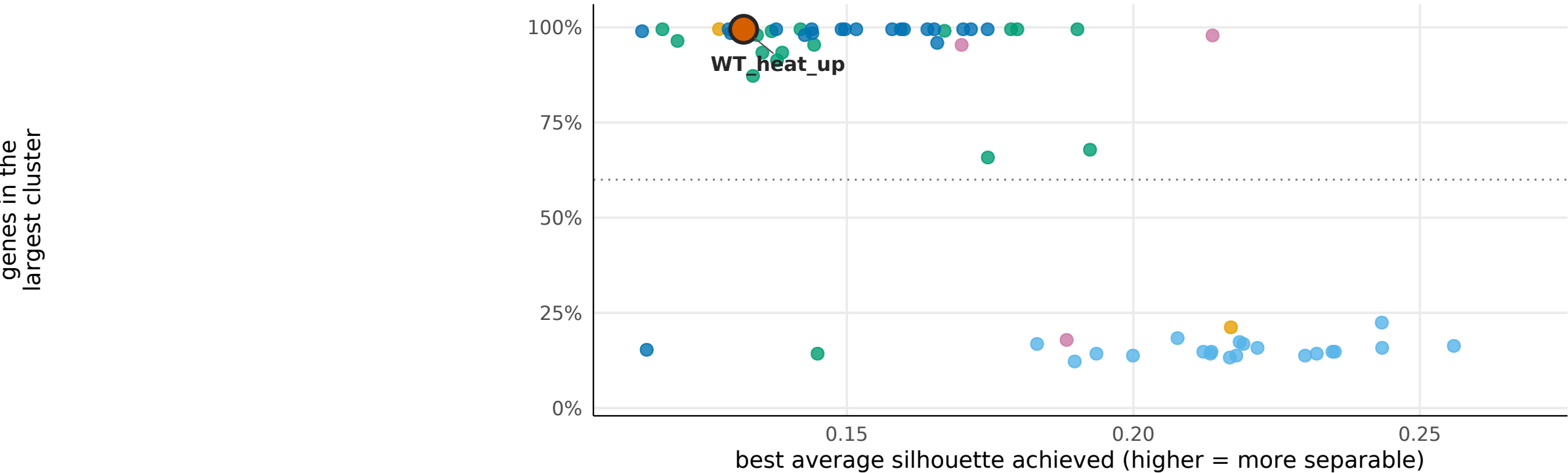
Is it one process, or many?

Mean pairwise Wang similarity, every set matched to the query's 196 genes. Against a null matched on annotation depth the set sits +1.38 SD (p 0.095). Against a uniform null it sits +4.39 SD, and that difference is the confound, not the biology: the gate admits genes carrying a median 9 BP terms where the expressed background carries 4.



Can it be split into modules?

Best of three linkages plus k-medoids, k = 2 to 15, by average silhouette. No gene-level partition is reported for this set: its best split leaves 99% of the genes in one group. A null matched on annotation depth is no better (95%), so being unsplittable is a property of well-annotated gene sets and not a finding about this one. The uniformly drawn null does separate (15%), because genes carrying one or two terms are semantic islands rather than because those sets have structure. Single-GO-term references, coherent by construction, also sit high (88%). Forcing k on any of these would draw an artefact.



- WT\_heat\_up (the set under test)
- one GO term and its offspring
- null matched on annotation depth
- null drawn uniformly
- curated lens
- sibling gate set